

VertKleen HCR and Descaler User Guide

DOCUMENT ID	MAS-VK-HCR-DESC-GUIDE
DISTRIBUTION REVISION	1.0
EFFECTIVE DATE	2026-07-24
SUPERSEDED STATUS	Current
OWNER	MASEST Consulting LLC
APPROVAL	Released for customer review
APPLICABLE SKU(S)	VK-HCR, VK-DESC, VK-CRS

TECHNICAL / LEGAL CLAIM REVIEW REQUIRED

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Approval scope: Distribution approval only; technical and legal claim approval is not asserted.

HEAT EXCHANGERS

Prepare your closed-loop heat exchanger as usual and circulate the VertKleen Descaler or HCR solution as follows.

- 0-50 gallon - 1 hour
- 500-200 gallon - 2 hours
- 200-500 gallon - 3 hours
- 500+ gallon - 4 hours

Monitor the solution each half hour using pH test strips (if strips are red, product is active. Otherwise, add more product).

Medium Descale - 25% VertKleen Descaler or HCR Solution

Light Descale - 10% VertKleen Descaler or HCR Solution

BOILER

Every boiler will have a varying degree of scale when it's time to clean it. Generally, adding 20% HCR to the water volume is where we recommend you start.

1. Close down boiler/components for closed-loop descaling in order to circulate cleaning solution
2. Drain and flush the boiler
3. Add HCR to the boiler then fill with water
4. Circulate for 3 hrs (6 for larger units)
5. Drain, flush, refill and you're done

To use the soak method, fill the boiler with one part HCR and one part water. Allow to soak for 2-3 hours.

With either cleaning method, monitor the solution each hour using pH test strips. Once acidity remains the fluid is active.

COOLING TOWERS

1. Blow down system
2. Drain sump
3. Fill system with VertKleen Descaler solution (see dilution below)
4. Allow solution to circulate for 3-4 hours
5. Drain, flush and refill make-up water.

Monitor the solution each half hour using pH test strips (if strips are red, product is active. Otherwise, add more product).

Descale - 5 gallons of VertKleen Descaler per ten ton volume
Light Descale - 3 gallons of VertKleen Descaler per ton volume

EVAPORATIVE COOLERS

VertKleen Descaler easily removes even heavy buildup from evaporative pads, water lines, screens and pumps.

Introduce Descaler at the needed strength to the water reservoir (see dilution rates below).

Run the pump for 20-30 minutes.

Drain sump, flush with water and re-fill with fresh water.

For pads that cannot be removed, dilute Descaler with 2 parts water, soak the pad for 10 minutes. Pads made with rigid materials can be lightly brushed to remove large scale debris that will loosen. Rinse well with water and re-install the pads.

For customers descaling potable water systems, MASEST's WaterSafe 60 carries NSF60 certification for use as a descaler and pH adjuster. It is also Kosher Pareve approved.

